Forecasting Baselines: Short-Term Point Forecasts for Estimated WEC Power

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This notebook builds simple short-term point-forecasting baselines for the estimated WEC power output prepared in the previous notebooks.

The input data used here is the 30-minute Leixões estimated WEC power time series. The sourcing and sea-state preparation are documented in wave data preparation, and the simplified WEC power estimation procedure is documented in WEC power estimation.

The main modelling target is `wec_power_norm_estimated`, which represents the estimated WEC power normalized by the nominal 250 kW rating used for the generic WEC proxy. This device-relative scale is useful for modelling because it keeps the target independent of larger rated-power or array-sizing assumptions.

For physical interpretation and downstream analysis, the normalized target is also converted to the corresponding 250 kW scale:

`wec_power_kw_estimated = wec_power_norm_estimated × 250 kW`

The main forecasting results in this notebook are reported and visualized on the normalized power scale. The 250 kW-scale true values, predictions, and errors are saved in the prediction output table for later interpretation in the uncertainty and storage-aware analysis notebooks.

The forecast target is not measured WEC power. It is also not a validated device model, hydrodynamic simulation, or wave-to-wire model. It is a simplified, transparent WEC power proxy derived from observed sea-state variables using the assumptions documented in the WEC power estimation notebook.

The purpose of this notebook is to create reproducible point-forecast baselines and residuals that can support the uncertainty and storage-aware subsequent analyses.

The benchmark uses a compact power-only autoregressive feature set. Wave-height, wave-period, and wave-power-flux variables are not used as predictors in the main benchmark, because the estimated WEC power target was itself constructed from sea-state assumptions. This avoids a forecast model that partly relearns the constructed power-matrix mapping instead of forecasting the estimated power series as a time-dependent signal.

The feature set is intentionally kept small: selected lag values, rolling means, and rolling standard deviations of the estimated power series. This keeps the benchmark interpretable and avoids turning the first forecasting notebook into a broad feature-search exercise.

The forecasting horizons are:

Horizon	Steps	Operational context
30 min	1	Grid-relevant: next balancing or dispatch interval
1 h	2	Grid/storage-relevant: near-term operational planning

Horizon	Steps	Operational context
2 h	4	Storage-relevant: short storage dispatch window
4 h	8	Grid/storage-relevant: upper short-term benchmark without external wave forecast inputs

Missing target values are not imputed. Lag, rolling, and future target values are created only within continuous valid segments of the original 30-minute time axis. This prevents observations separated by data gaps from being treated as adjacent 30-minute observations.

Evaluation uses chronological rolling-origin folds. Each fold contains a training block, a calibration/validation block, and a final test block. Using several rolling-origin test folds gives multiple performance estimates across different time periods, which helps assess the stability of the forecasting baselines rather than relying on a single final split.

The notebook compares a small set of interpretable baseline models:

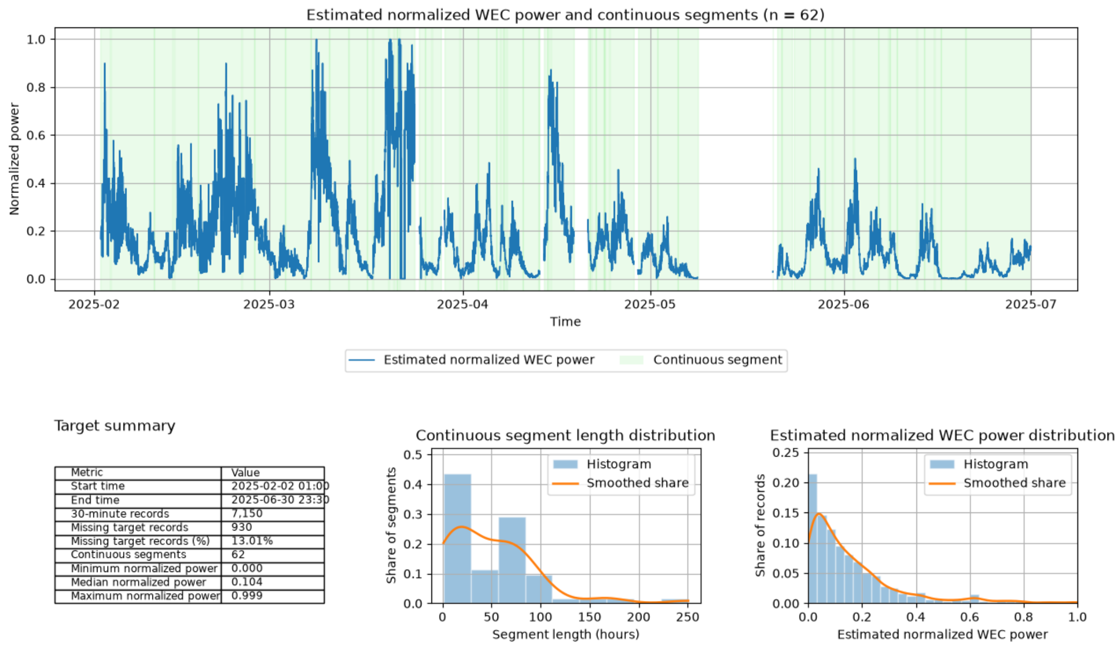
- Persistence
- Rolling mean
- Ridge regression
- Random forest regression

Performance is reported using MAE, RMSE, and skill scores relative to persistence.

Target availability and continuous segments

Before creating forecasting features, the estimated power target is checked on the original 30-minute time axis.

Missing target values are kept as gaps. Continuous valid segments are identified so that lag, rolling, and future-target features can later be created only within uninterrupted parts of the series.



Metric	Value
Start time	2025-02-02 01:00
End time	2025-06-30 23:30
30-minute records	7,150
Missing target records	930
Missing target records (%)	13.01%
Continuous segments	62
Minimum normalized power	0.000
Median normalized power	0.104
Maximum normalized power	0.999

```
Saved target availability figure to:
../outputs/figures/notebook_03/target_availability_and_segments.png
```

The target series contains 62 continuous non-missing segments, with about 13% missing target records on the original 30-minute time axis. Estimated normalized WEC power is concentrated at low-to-moderate output levels, with fewer near-rated values.

The following forecasting features are therefore created within each continuous segment only, so lagged and rolling predictors do not cross missing-data gaps.

Segment-safe forecasting features

Forecasting features are created from the estimated normalized WEC power series only.

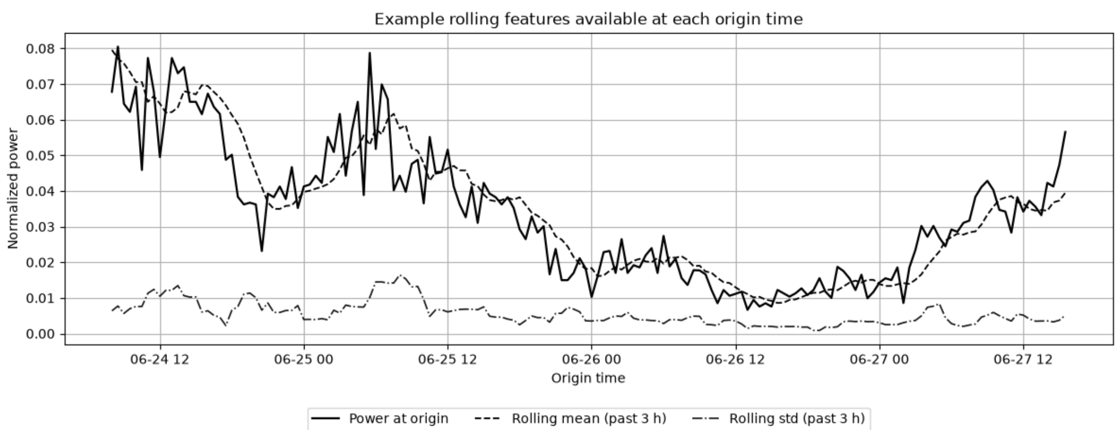
For each forecast horizon, the regression target is the future estimated normalized WEC power, `y_target_norm`, evaluated at the corresponding `target_time`.

The same compact autoregressive feature set is used for all horizons:

- `power_lag_1`
- `power_lag_2`
- `power_lag_4`
- `power_lag_8`
- `power_lag_24`
- `power_roll_mean_3`
- `power_roll_mean_6`
- `power_roll_mean_12`
- `power_roll_std_6`
- `power_roll_std_12`

Lag and rolling features are computed within each continuous segment. Rolling features use past values only. Future targets are also created within each segment, so forecast examples do not cross missing-data gaps.

horizon	horizon_steps	horizon_hours	supervised_rows	share_of_available_target_records
30 min	1	0.5	4866	78.23
1 h	2	1	4819	77.48
2 h	4	2	4727	76
4 h	8	4	4545	73.07



Rolling-origin folds

The supervised rows are split chronologically into 5 rolling-origin folds.

Each fold uses an expanding chronological prefix of the supervised record. Within each fold, the split is approximately 70% training, 15% calibration/validation, and 15% test. The later folds use longer prefixes, so they include more training and evaluation samples.

The calibration/validation block is kept separate because the same fold structure will later support prediction-interval calibration in the uncertainty analysis.

horizon_steps	horizon_label	fold_id	train	calibration	test
1	30 min	1	2724 (70.0%)	583 (15.0%)	585 (15.0%)
1	30 min	2	2894 (70.0%)	620 (15.0%)	621 (15.0%)
1	30 min	3	3065 (70.0%)	656 (15.0%)	658 (15.0%)
1	30 min	4	3235 (70.0%)	693 (15.0%)	694 (15.0%)
1	30 min	5	3406 (70.0%)	729 (15.0%)	731 (15.0%)
2	1 h	1	2698 (70.0%)	578 (15.0%)	579 (15.0%)
2	1 h	2	2867 (70.0%)	614 (15.0%)	615 (15.0%)
2	1 h	3	3035 (70.0%)	650 (15.0%)	652 (15.0%)
2	1 h	4	3204 (70.0%)	686 (15.0%)	688 (15.0%)
2	1 h	5	3373 (70.0%)	722 (15.0%)	724 (15.0%)

horizon_steps	horizon_label	fold_id	train	calibration	test
4	2 h	1	2646 (70.0%)	567 (15.0%)	568 (15.0%)
4	2 h	2	2811 (70.0%)	602 (15.0%)	604 (15.0%)
4	2 h	3	2977 (70.0%)	638 (15.0%)	639 (15.0%)
4	2 h	4	3143 (70.0%)	673 (15.0%)	674 (15.0%)
4	2 h	5	3308 (70.0%)	709 (15.0%)	710 (15.0%)
8	4 h	1	2545 (70.0%)	545 (15.0%)	546 (15.0%)
8	4 h	2	2704 (70.0%)	579 (15.0%)	580 (15.0%)
8	4 h	3	2863 (70.0%)	613 (15.0%)	614 (15.0%)
8	4 h	4	3021 (70.0%)	647 (15.0%)	649 (15.0%)
8	4 h	5	3181 (70.0%)	681 (15.0%)	683 (15.0%)

```
Saved fold definitions to: ../data/processed/folds/forecast_folds.parquet
```

Forecasting models

The benchmark uses four simple point-forecasting models:

- **Persistence** — predicts the future value using the current estimated power at the forecast origin. No hyperparameters.
- **Rolling mean** — predicts the future value using a recent rolling mean of estimated power, with the rolling window selected from $w \in \{3, 6, 12\}$ 30-minute steps.
- **Ridge regression** — regularized linear autoregressive model implemented with scikit-learn, with regularization strength selected from $\alpha \in \{0.1, 1.0, 10.0\}$.
- **Random forest regression** — non-linear tree-based benchmark implemented with scikit-learn, with a small grid over tree depth and leaf size: $d_{\text{max}} \in \{6, 12, \text{None}\}$ and $n_{\text{min leaf}} \in \{3, 5, 10\}$.

Ridge regression uses a `StandardScaler` inside a scikit-learn pipeline, fitted only on the training block of each fold. The baseline models and random forest do not require feature scaling.

Hyperparameter selection uses a small manual grid search within each fold: models are fitted on the training block, candidate settings are compared on the calibration/validation block, and final performance is reported on the test block.

Fold-wise model fitting and prediction

For each horizon and fold, candidate model settings are fitted on the training block and compared on the calibration/validation block. The selected setting is then used to generate predictions for the calibration and test blocks.

The test block is not used for model fitting or hyperparameter selection.

The saved prediction table keeps the selected point forecasts and residuals for downstream uncertainty analysis. Fitted fold-specific models are also saved so the selected forecasting objects can be inspected or reused later.

```
Fitting forecast models: 100%|          | 80/80 [08:03<00:00,  6.05s/it,
horizon=4 h, fold=5, model=RandomForest]
```

horizon_steps	Horizon	Model setting	Folds	Calib. RMSE	Calib. MAE
1	30 min	Ridge (alpha=0.1)	4	0.031023	0.0210353
1	30 min	RollingMean (feature=power_roll_mean_3)	5	0.0312866	0.0198615
1	30 min	RandomForest (max_depth=6, min_samples_leaf=5)	4	0.032263	0.0216087
1	30 min	Ridge (alpha=10.0)	1	0.0333056	0.0231719
1	30 min	RandomForest (max_depth=6, min_samples_leaf=3)	1	0.0335616	0.0228114
1	30 min	Persistence	5	0.0344685	0.019806
2	1 h	Ridge (alpha=0.1)	2	0.0280165	0.0200535
2	1 h	RandomForest (max_depth=6, min_samples_leaf=5)	4	0.0332452	0.022801
2	1 h	RollingMean (feature=power_roll_mean_3)	5	0.0332939	0.0214555
2	1 h	Ridge (alpha=10.0)	3	0.0365718	0.0251558
2	1 h	Persistence	5	0.0365909	0.0218935
2	1 h	RandomForest (max_depth=6, min_samples_leaf=3)	1	0.0388285	0.024901
4	2 h	RandomForest (max_depth=12, min_samples_leaf=3)	1	0.0315008	0.0235138
4	2 h	Ridge (alpha=0.1)	2	0.0331265	0.0237499
4	2 h	RandomForest (max_depth=None, min_samples_leaf=3)	1	0.0385057	0.0260482
4	2 h	RandomForest (max_depth=6, min_samples_leaf=3)	1	0.039017	0.0274737
4	2 h	RollingMean (feature=power_roll_mean_3)	5	0.0391015	0.0254323
4	2 h	Persistence	5	0.0392757	0.0243821
4	2 h	Ridge (alpha=10.0)	3	0.0422863	0.0294757
4	2 h	RandomForest (max_depth=None, min_samples_leaf=5)	2	0.0443432	0.0297228
8	4 h	Ridge (alpha=1.0)	1	0.0470631	0.033967
8	4 h	RollingMean (feature=power_roll_mean_3)	5	0.0481108	0.0324157
8	4 h	Persistence	5	0.0481373	0.0319743
8	4 h	Ridge (alpha=10.0)	4	0.0485587	0.0355872
8	4 h	RandomForest (max_depth=6, min_samples_leaf=3)	2	0.0511552	0.0357079
8	4 h	RandomForest (max_depth=None, min_samples_leaf=5)	1	0.0580243	0.037908
8	4 h	RandomForest (max_depth=6, min_samples_leaf=5)	2	0.0580859	0.0386124

horizon_steps	Horizon	Model	Tune time (s)	Fit time (s)	Test pred. time (s)
1	30 min	Persistence	0.00828986	3.0658e-06	0.000131944
1	30 min	RollingMean	0.0120804	3.1112e-06	0.000142865
1	30 min	Ridge	0.0559844	0.00902434	0.00372598
1	30 min	RandomForest	22.7416	1.95059	0.195939
2	1 h	Persistence	0.00709714	3.8448e-06	0.00014441
2	1 h	RollingMean	0.0119146	2.874e-06	0.000115681
2	1 h	Ridge	0.0516085	0.00845967	0.00351877
2	1 h	RandomForest	22.7173	1.82353	0.193201
4	2 h	Persistence	0.00656106	3.3548e-06	0.000162199
4	2 h	RollingMean	0.0123179	3.1242e-06	9.2382e-05
4	2 h	Ridge	0.0472068	0.00745153	0.00321063
4	2 h	RandomForest	20.4424	2.27005	0.309291
8	4 h	Persistence	0.00872855	4.1592e-06	0.000143411
8	4 h	RollingMean	0.0142514	3.4568e-06	9.75054e-05
8	4 h	Ridge	0.0489788	0.00789193	0.00441131
8	4 h	RandomForest	19.1528	1.57721	0.319706

```
Saved selected predictions and residuals to:
../outputs/notebook_03/forecast_predictions.parquet
Saved model-selection table to:
../outputs/tables/notebook_03/model_selection.csv
Saved model-runtime table to: ../outputs/tables/notebook_03/model_runtime.csv
Saved model-selection summary to:
../outputs/tables/notebook_03/model_selection_summary.csv
Saved model-runtime summary to:
../outputs/tables/notebook_03/model_runtime_summary.csv
Saved selected fitted models to: ../outputs/models/notebook_03
```

Test-set forecasting performance

Final forecasting performance is evaluated on the test block of each rolling-origin fold.

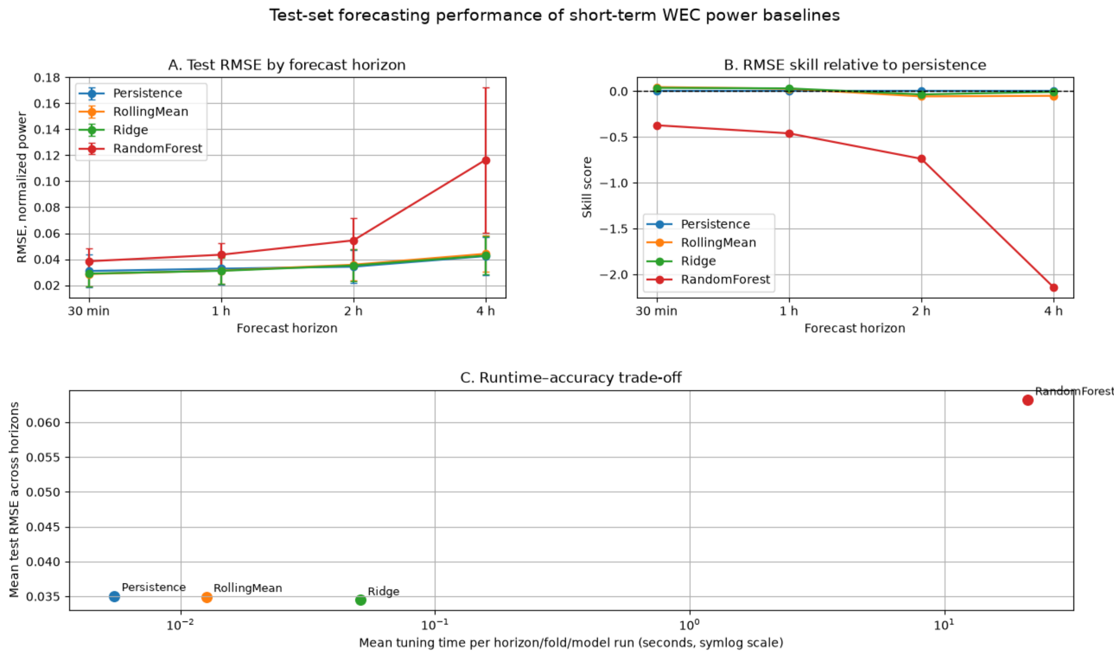
The metrics below use only the selected tuned model predictions. The calibration/validation block is used for model selection, while the test block is reserved for final evaluation.

Errors are reported on the normalized WEC power scale and on the corresponding 250 kW interpretation scale.

Skill scores are computed relative to persistence:

$$\text{Skill} = 1 - \frac{\text{model error}}{\text{persistence error}}$$

Positive skill means the model improves over persistence, zero means equal performance, and negative skill means worse performance than persistence.




```
Saved fold-level test metrics to:
../outputs/tables/notebook_03/forecast_metrics_by_fold.csv
Saved summary test metrics to:
../outputs/tables/notebook_03/forecast_metrics_summary.csv
Saved test performance figure to:
../outputs/figures/notebook_03/test_forecasting_performance.png
```

The test-set results show that the simple autoregressive baselines are strong for this short-term forecasting task. Ridge regression gives the most balanced performance overall, with low RMSE, competitive or positive skill relative to persistence, and low computational cost.

Persistence and the rolling-mean baseline remain competitive, especially at shorter horizons, which is expected for an autocorrelated estimated WEC power series. The random forest model is substantially slower and does not improve test performance in this setup. This suggests that, for the current power-only feature set and limited short-term horizons, model simplicity is preferable to added non-linear complexity.

As expected, forecast error generally increases with horizon, reflecting the increasing difficulty of predicting estimated WEC power further ahead without external wave-forecast inputs.

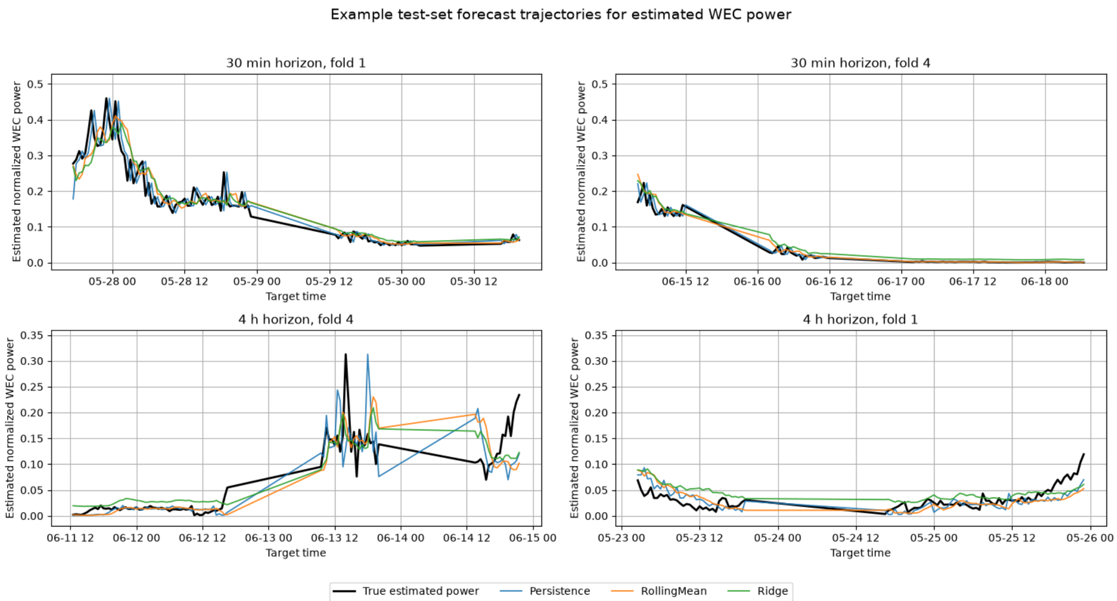
Example forecast trajectories

The aggregate test metrics show overall performance, but they do not show how the forecasts behave through time.

This section visualizes example test-set forecast trajectories for the shortest and longest forecast horizons. Forecasts are aligned by `target_time`, so each prediction is compared with the realized estimated WEC power at the time being forecast.

Only the main interpretable models are shown: persistence, rolling mean, and ridge regression. The random forest is omitted from this trajectory plot because the previous performance comparison showed that it is slower and not competitive for this setup.

The plotted windows are selected automatically from the final test fold by choosing periods with relatively high variation in the true target series.




```
Saved example forecast trajectory figure to:
../outputs/figures/notebook_03/example_forecast_trajectories_2x2.png
```

The example trajectories show that the selected baselines capture the broad evolution of estimated WEC power reasonably well, especially at the 30 min horizon. Short-term fluctuations and abrupt changes are harder to follow, which is expected for a wave-driven power signal with strong variability.

At the 4 h horizon, forecasts become smoother and show more lag around changing conditions. This is consistent with the aggregate metrics: longer forecast horizons are more difficult without external wave-forecast inputs.

Overall, the trajectories support the main test-set result: simple autoregressive baselines, especially ridge regression and rolling-mean forecasting, provide intuitive and competitive short-term forecasts for the simplified estimated WEC power target.

Residual diagnostics

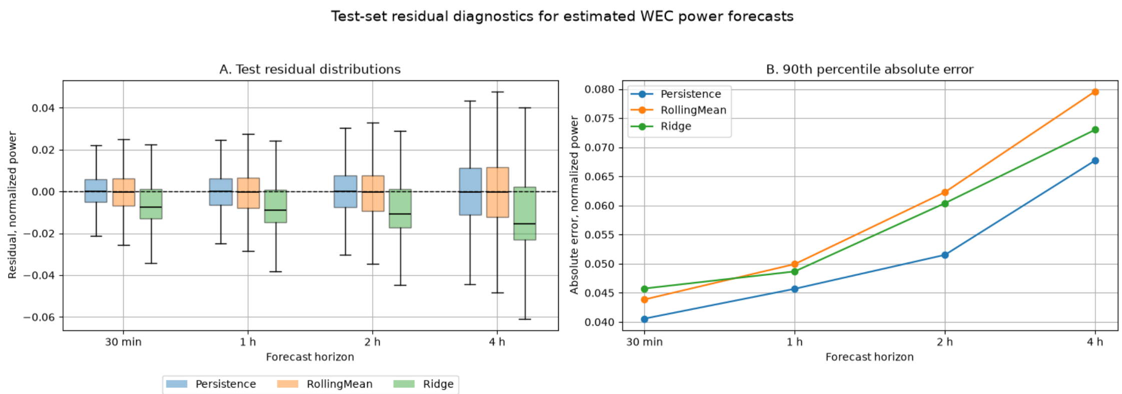
The point-forecast metrics show average test performance, but the residuals show how forecast errors are distributed.

This section examines test-set residuals for the main interpretable baselines. Residuals are defined as:

$$e_t = y_t - \hat{y}_t$$

Positive residuals mean the model under-predicted the estimated WEC power, while negative residuals mean it over-predicted.

These residuals are especially important for the next notebook, where forecast uncertainty and prediction intervals are evaluated.




```
Saved residual diagnostic summary to:
../outputs/tables/notebook_03/residual_diagnostics_summary.csv
Saved residual diagnostic figure to:
../outputs/figures/notebook_03/residual_diagnostics.png
```

The residual diagnostics show that test residuals are generally centered close to zero, indicating no large systematic bias for the main baselines. However, the residual spread increases with forecast horizon, which is consistent with the increasing difficulty of longer-ahead WEC power forecasting.

Persistence has the lowest 90th percentile absolute error in this diagnostic view, especially at longer horizons. Ridge remains a competitive low-cost ML baseline, but its residuals show a slightly stronger tendency toward negative errors at longer horizons, meaning it sometimes over-predicts the estimated WEC power.

These results reinforce the importance of evaluating both average metrics and residual behavior. For the next uncertainty notebook, the residual spread and horizon-dependent error growth provide a direct motivation for calibrated prediction intervals rather than relying only on point forecasts.

Summary

This notebook built short-term point-forecasting baselines for the estimated WEC power time series prepared in the previous notebooks.

The forecasting task used the normalized estimated WEC power target as the main modelling and visualization scale. Corresponding 250 kW-scale true values, predictions, and errors were also computed and saved for downstream physical interpretation. Missing target values were not imputed. Lag, rolling, and future-target features were created only within continuous valid segments, preventing artificial continuity across data gaps.

A compact power-only autoregressive feature set was used for all horizons. Sea-state variables such as wave height, wave period, and wave-power flux were intentionally excluded from the main benchmark because the estimated WEC power target was itself derived from sea-state assumptions.

The evaluation used chronological rolling-origin folds with separate training, calibration/validation, and test blocks. Hyperparameters were selected on the calibration block, while final performance was evaluated only on the test block.

The main findings were:

- Forecast error increased with horizon, as expected for longer-ahead WEC power forecasting without external wave-forecast inputs.
- Persistence and rolling-mean baselines were strong, especially at short horizons, reflecting the autocorrelation of the estimated WEC power series.
- Ridge regression provided the most balanced ML baseline, combining competitive test performance with low computational cost.
- Random forest regression was substantially slower and did not improve test performance in this compact power-only setup.
- Residual diagnostics showed increasing error spread with horizon, motivating calibrated prediction intervals in the next notebook.

The saved prediction and residual outputs from this notebook provide the point-forecast basis for the following uncertainty-analysis notebook.